# Gender Differences in the Returns to Double Majoring: A Case of Signaling?

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**Abstract**

Over 15 percent of college graduates in the United States graduate with more than one major. While much research exists on the returns to different individual majors, less is known about the causal effects of double majoring. This study provides novel estimates on the returns to those who choose to double major as an undergraduate. We improve on previous studies that rely on controls for observable characteristics by addressing selection concerns in two notable ways. First, we control for institution-specific differences that may influence both the decision to double major and subsequent earnings using proprietary data by including institution fixed effects. Second, we adopt a partial identification approach to address non-random selection into double majoring within institutions. Results broadly align with estimates from prior studies at the aggregate level, but reveal notable gender differences not previously detected. Women experience an earnings return to double majoring of approximately 5 percent, while the return for men is statistically negligible. Our analysis suggests that this discrepancy is consistent with double majors serving as a signaling effect in the labor market, which may help offset the gender pay gap.

**Keyword:** double majoring; returns to education; gender wage gap; labor market signaling; human capital; college major choice

**JEL Classification:** I26; J16; J31; I23; J24

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Data Availability — This paper uses confidential data from the National Survey of College Graduates maintained by the Federal Statistical Research Data Centers (FSRDC). The data can be obtained by submitting a request through the FSRDC. The authors are willing to assist in the process.

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# 1 Introduction

A substantial body of research indicates that there are sizable returns to education, and that the returns to higher education in particular have been increasing in recent years, both in absolute and relative terms (Goldin and Katz, 2007; Binder and Bound, 2019; Deming, 2022). This trend has coincided with a significant increase in U.S. college attainment rates over the past century (Lovenheim and Turner, 2019). Several studies have looked at the returns to different college majors, generally finding large differences in earnings returns across fields.[1] However, much less is known about the returns to double majoring, even though over 15 percent of recent U.S. college graduates have a degree with multiple majors.

This paper evaluates the returns to double majoring on several later life outcomes, including earnings, employment, marital status, fertility, and graduate education. We use restricted-access data from six waves of the National Survey of College Graduates (NSCG) spanning from 2010 to 2021. These surveys provide a repeated cross-sectional random sample of U.S college graduates and contain detailed information on labor market outcomes, educational attainment, and demographic characteristics. Furthermore, the restricted-access NSCG data contain college and university codes that allow us to identify individuals who graduated from the same institution.

Measuring the causal effects of double majoring is challenging due to nonrandom student selection into double majoring. A student's decision to double major is likely influenced by unobservable factors, leading to omitted variable bias concerns when attempting to measure the causal relationship between double majoring and later life outcomes, even when controlling for observed characteristics. First, students sort into colleges based on unobserved factors such as ability, preferences, and ambition, all of which affect later life outcomes. This poses a challenge to causal inference, since policies and norms around double majoring likely vary across institutions, leading unobserved factors to influence both the propensity to double major and future outcomes. Second, even within institutions, there is likely sorting into decisions to double major due to personal characteristics that also affect later life outcomes. For example, students with greater ambition may be more likely to pursue a double major, and ambition likely affects future outcomes independently of double majoring.

We tackle these sorting concerns using multiple strategies. First, to address concerns about nonrandom sorting across colleges and universities, we include institution fixed effects in our estimations. Second, to address concerns about nonrandom sorting within institutions,

1. Altonji, Blom, and Meghir (2012) and Lovenheim and Smith (2022) provide reviews of the literature on returns to college majors.

we estimate bounds on the causal effects of double majoring using the sensitivity analysis introduced by Diegert, Masten, and Poirier (2023). Intuitively, this method assesses the robustness of estimates to selection on unobserved characteristics by using information on the stability of estimated coefficients to the inclusion of observed control variables. This method is related to the popular sensitivity analysis by Oster (2019). We discuss our motivation for choosing Diegert, Masten, and Poirier (2023) as our preferred specification in Section 2.[2]

Results indicate significant gender differences in the labor market returns to double majoring. The earnings return for women is approximately 5 percent, in line with prior literature. However, the return for men is considerably smaller and statistically indistinguishable from zero. Notably, double majoring does not statistically affect employment for either gender, suggesting that earnings results are not driven by selection into employment. Additionally, we find that double majoring increases women's propensity to obtain graduate degrees at almost twice the rate for men. Double majoring does not perceptibly increase the probability for job-degree match for men or women, defined by the NSCG as having a "principal job closely related to highest degree." Double majoring decreases the propensity to marry for both men and women and decreases the propensity to have children in the household for women, but not men.[3] We explore the possibility that results are driven by compositional differences in the majors that men and women tend to pursue. Namely, we explore whether results appear to be driven by gender differences in the propensity to diversify double majors in terms of field or gender differences in the specific field compositions of double majors. We do not find evidence that differences in major composition drive the estimated gender differences in returns to double majoring.

Conceptually, the returns to double majoring could be explained by a number of competing theories, the foremost among them, the acquisition of human capital and the role of a double major as a labor market signal. We assess that the returns to a double major for women are most consistent with a model of labor market signaling. Specifically, women may be using the double major as a signal of their labor market ambition, both practically, in terms of career orientation, and academically, in terms of the propensity to pursue education beyond a bachelor's degree. In both cases, women with double majors appear to show a decreased propensity to have children, which is especially salient. Pure human capital theory appears to be less relevant, especially in light of the null effects for men, who arguably also attain additional human capital by acquiring a double major but do not show a statistically positive return.

2. In the Online Appendix, we show that our findings are robust to using Oster (2019)'s method.

3. Note that these fertility effects could indicate that women who double major are having fewer children or that they are delaying childbearing.

This paper contributes to a growing literature investigating the causal effects of college majors on post-college outcomes. Several papers investigate earnings returns to single majors, generally finding significant heterogeneity in returns across majors or fields of study (Arcidiacono, 2004; Hamermesh and Donald, 2008; Webber, 2014; Hastings, Neilson, and Zimmerman, 2013; Kirkeboen, Leuven, and Mogstad, 2016). More recent papers build on previous work to provide nuance to the subject by investigating the effects of students' majors on earnings growth and variability, mechanisms behind the returns, and interaction effects between degree choice and university selectivity (Andrews et al., 2022; Bleemer and Mehta, 2022; Britton et al., 2022; Choi et al., 2024).

In comparison to the literature on single majors, much less has been written about the returns to a double major. Using data on students at Northwestern University, Zafar (2012) finds evidence that students are strategic in their double major choices and choose pairs of majors that differ in terms of difficulty, likelihood of completion, and chances of finding a job after graduation. In line with this, Hanks et al. (2024) analyze data from the American Community Survey and find that a double major may be used as a diversification tool to hedge against income risk by trading off lower average earnings for more consistency in their earnings. They focus on the role of double major in affecting variance in earnings rather than levels in earnings. Existing studies looking at the relationship between double majoring and earnings levels primarily rely on ordinary least squares (OLS) estimation with control variables. Del Rossi and Hersch (2008) take this approach, finding that having a second major is related to 2.3 percent higher earnings on average, compared to having a single major. Similarly, Hemelt (2010) finds a 3.2 percent average relationship, and Del Rossi and Hersch (2016) estimate an imprecise zero relationship.[4] Zhu and Zhang (2021) uses the Baccalaureate and Beyond data and implement a propensity score weighting approach. They find that double majors initially earn less than their single major peers with the same higher-paying major, but earnings converge four years after graduation. They also find that students who double major have higher graduation rates, work more hours after graduation, and are more likely to attend graduate school than their single major peers.

This study builds on prior work in several ways. First, we incorporate novel information from restricted-use data to address selection across institutions by incorporating institution fixed effects. Second, we use partial identification tools to assess the relevance of selection on unobservables within institutions. These additions allow us to estimate causal effects of double majors that address concerns about selection on unobservable characteristics. Finally,

4. Del Rossi and Hersch, 2008, Hemelt, 2010, and Del Rossi and Hersch, 2016 use the NSCG for their analysis as well. Key differences between these papers and ours are that i) each paper uses only a single wave of the survey, and ii) these studies do not have access to restricted-use institutional identifiers.

this study analyzes a larger sample size relative to prior work, allowing us to better examine heterogeneity by gender and provide a nuanced exploration of the mechanisms underlying our results.

Additionally, our findings speak to the literature studying the relationship between college major choice and the gender wage gap.[5] Several studies have documented gender differences in major choice, as well as examining factors driving these patterns (Dickson, 2010; Turner and Bowen, 1999; Kugler, Tinsley, and Ukhaneva, 2021; Zafar, 2013; Sloane, Hurst, and Black, 2021; Speer, 2017; Porter and Serra, 2020; Wiswall and Zafar, 2015). These differences in major choice, in turn, have significant implications for gender gaps in the labor market given the high degree of variation in returns to different majors. More recently, Imberman et al. (2024) have documented that in addition to differences in the distribution of majors by gender, there are sizable gender differences in major-specific returns that contribute to the gender earnings gap. Lepage, Li, and Zafar (2025) examine mechanisms behind gender differences in major choices. Their findings suggest that labor market signaling plays an important role, with women selecting majors in anticipation of potential labor market discrimination. While prior studies have largely focused on gender differences in single majors and the implications of this for the gender earnings gap, we provide novel findings on the role of double majoring in affecting gender gaps in labor market outcomes.

# 2 Empirical Strategy

## 2.1 Overview

We estimate a series of regressions of the following form to assess the effects of double majoring on later life outcomes:

$$Y_{isnat} = \beta D_{isnat} + X'_{isnat}\gamma + \eta_a + \pi_s + \tau_t + \theta_n + \left(W'_{isnat}\alpha + \epsilon_{isnat}\right) \tag{1}$$

where $Y_{isnat}$ denotes the outcome of interest for individual $i$ born in state $s$ graduating from institution $n$ who is age $a$ in survey year $t$. The outcome is a function of $D_{isnat}$, an indicator variable that takes a value of one if individual $i$ has a double major and zero otherwise. Our coefficient of interest is $\beta$, which represents the causal returns to double majoring under some assumptions. It captures the effect of an ideal thought experiment in which we randomly assign a treatment group of students to obtain a double major, while the control group

5. For a review of studies on the gender wage gap, see Goldin, Kerr, and Olivetti (2022) and Blau and Kahn (2017).

is assigned to not complete requirements for a double major. However, to assert that an estimate of $\beta$ is causal is challenging in a real-world setting because individuals are not randomly assigned to single versus double majors. Among those who receive a bachelor's degree, the decision to double major is likely driven by a combination of observed and unobserved factors. To address sorting on observable characteristics, we control for a vector of variables, $X_{isnat}$, which includes indicators for race and ethnicity, gender, and parental education. Additionally, we control for a set of indicators for age, state of birth, and year fixed effects, denoted by $\eta_a$, $\pi_s$, and $\tau_t$, respectively. We discuss the composite error term, $\left(W'_{isnat}\alpha + \epsilon_{isnat}\right)$, in the next section.

To obtain an unbiased estimate of $\beta$, we need to address sorting on unobservable characteristics on two dimensions: First, students sort into colleges and universities based on factors such as ability and preferences that likely also affect later life outcomes; moreover, policies around double majoring vary significantly across institutions. Second, even within institutions, there is likely sorting into decisions to double major due to personal characteristics that also affect later life outcomes. There is likely an overlap in the unobservable factors that lead to sorting along these two dimensions. Therefore, we control for a likely important component of unobserved ability and preferences by including institution fixed effects, $\theta_n$. Institution fixed effects control for both the possibility that individuals with different propensities to double major (e.g., due to different levels of underlying ambition) sort nonrandomly across institutions and for the possibility that requirements and norms surrounding double majors differ across institutions. Next, to assess the role of any remaining within-institution sorting on unobservable characteristics, we construct bounds on the causal effects of double majoring using the sensitivity analysis method of Diegert, Masten, and Poirier (2023), which builds on the popular Oster (2019) method. Altogether, our empirical strategy allows us to conduct statistical inference on the causal returns to double majoring on later life outcomes by accounting for the role of selection on unobservable characteristics both across and within institutions.

## 2.2 Addressing within-institution sorting on unobservable characteristics

The error term in Equation 1 is a composite of a traditional independent and identically distributed error, $\epsilon_{isnat}$, and an index, $W'_{isnat}\alpha$, representing the influence of all other factors (such as within-institution differences in ability) that affect the outcome but are unobserved. The OLS estimate, $\hat{\beta}$, will be equal to the causal effect, $\beta$, only if $W_{isnat}$ is uncorrelated with selection into double majoring, which may be judged as unlikely a priori. However, depending

on the magnitude of selection on unobservables within institutions, the bias resulting from these omitted variables may or may not be substantial. To formally assess this, we apply the sensitivity analysis developed in Diegert, Masten, and Poirier (2023).

Consider the following "selection equation":

$$D_{isnat} = X'_{isnat}\delta + \kappa_a + \lambda_s + \mu_t + \psi_n + \left(W'_{isnat}\zeta + \nu_{isnat}\right) \tag{2}$$

which determines the choice of a double major, conditional on the same set of variables as in Equation 1. The sensitivity analysis by Diegert, Masten, and Poirier (2023) is based on the "relative variation in selection equation indices", denoted as $r_D$ below. Since $r_D$ is a non-negative ratio depending on an unobserved index, $W'_{isnat}\zeta$, sensitivity analysis can be performed by assuming that there exists some known $\bar{r}_D$ such that

$$0 \leq r_D = \frac{\sqrt{var(W'_{isnat}\zeta)}}{\sqrt{var(X'_{isnat}\delta)}} \leq \bar{r}_D.$$

$r_D$ can be interpreted as a ratio of standard deviations that measures the degree of unobserved selection due to $W_{isnat}$ relative to the observed selection due to $X_{isnat}$ for the selection into double majoring described by Equation 2. By assuming a value for $\bar{r}_D$, we can bound the true value of $\beta$ in Equation 1. Below, we will use this approach for different values of $\bar{r}_D$ to show the entire identified set, that is, the bounds on $\beta$ derived at each possible assumed value of $\bar{r}_D$. Alternatively, we also assess the robustness of our estimates of $\beta$ by calculating the value of $\bar{r}_D$ such that the bounds just include zero. This "breakdown point," denoted by $\bar{r}_D^{bp}(0)$, tells us how much selection on unobservables there would have to be relative to selection on observables in order to overturn our findings based on $\hat{\beta}$.

The general procedure to obtain bounds using this approach is similar to the popular Oster (2019) sensitivity analysis. However, Oster (2019) uses a ratio of regression coefficients from Equation 1 (the so-called Oster's $\delta$). A key implication of using this ratio is that, to compute those regression coefficients free of bias, Equation 1 has to be residualized, that is, the observed covariates have to be partialled out in a first step.[6] Diegert, Masten, and Poirier (2023) show that this residualization would in general result in sensitivity analysis that is not accurate, and this is something that does not happen with their sensitivity analysis based on $r_D$.[7] This is why we employ sensitivity analysis from Diegert, Masten, and Poirier (2023)

6. More precisely, the residualization is necessary if the observed variables in $X'_{isnat}$ are correlated to the unobserved variables $W'_{isnat}$, which is expected to be the case in most applications.

7. Formally, Diegert, Masten, and Poirier (2023) present a theoretical setting where the two ratios measuring the importance of unobserved covariates can be compared, and show that Oster (2019)'s ratio based on

in our preferred specification while showing in the Online Appendix that we obtain similar results using the Oster (2019) method.

With sensitivity analysis that benchmarks the possible selection on unobservables to the selection based on observables, it is important to pay attention to the observables that provide the benchmark. For instance, a benchmark of equal selection—that the unobservables are as important for selection as the observables—is not necessarily desirable if the observable covariates are deemed strong determinants of selection. Diegert, Masten, and Poirier (2023) suggest partitioning the observed variables into two subsets. The first set contains variables that are not part of the benchmarking of unobservables. We choose the complete set of fixed effects (year, institution, and state of birth) plus age indicators as part of this first set. The second set ($X_{isnat}$) contains parents' education and race and ethnicity indicators. Therefore, our sensitivity analysis based on $r_D$ benchmarks potential selection on unobservables to the observed variables "parents' education and race and ethnicity indicators" conditional on the set of fixed effects plus age indicators.

# 3 Data

Data for this project are drawn from the 2010, 2013, 2015, 2017, 2019, and 2021 waves of the National Survey of College Graduates (NSCG), conducted by the U.S. Census Bureau (*National Survey of College Graduates,* 2010, 2013, 2015, 2017, 2019, 2021). These surveys form a repeated cross-sectional random sample of non-institutionalized U.S. college graduates who hold at least a bachelor's degree, live in the United States, and are younger than 76 years. Individuals are selected to participate in the NSCG from the sample of American Community Survey participants, and the NSCG gathers information on various labor market outcomes, additional educational attainment, and demographic characteristics. This paper uses restricted-use NSCG data, which includes college and university codes, enabling us to identify individuals who graduated from the same institutions. A key benefit of the NSCG data set is that it contains detailed information about each college major earned by a student. We classify individuals as having a double major if they list a second major for their first bachelor's degree.

Table 1 displays summary statistics by gender for the pooled sample of 527,599 observations. Overall, students with double majors are fairly similar to those with single majors in terms of racial and ethnic composition and parental education for both men and women.

the preliminary residualization of Equation 1 does not have some desirable properties, such as its distribution being centered at one when the true ratio implies "equal selection" in terms of observed and unobserved covariates.

On average, double majors are older than their single major counterparts and received their first bachelor's degree earlier as a result. They are also more likely to have obtained their bachelor's degree from a liberal arts college or comprehensive college and less likely to have obtained their degree from a research university.[8,9] Our outcome variables of interest include labor market indicators, earnings and employment status, as well as a number of other outcomes that are of interest on their own but also help illuminate potential mechanisms: marital status, number of children in the household, job match, and graduate education.[10] Notably, summary statistics indicate that double majors have fewer children, higher levels of graduate education, and are less likely to be employed than single majors.

Figure 1 shows trends in double majoring over time. Rates of double majoring have been consistently about 5 percentage points higher for women than for men in all the birth cohorts we observe in our sample. The overall rate of double majoring for later birth cohorts has declined by about half for both men and women from the rates of the cohorts for those born 1935-1940. Double majoring rates became fairly constant at around 15 percent starting with birth cohorts in the 1970s and continuing through the most recent birth cohorts in the 1990s.

Next, Table 2 displays the distributions of single and double majors by area of study, separately for men and women. Following Del Rossi and Hersch, 2008, we group majors into five area of study categories: arts/social science, business (including economics), education, engineering, and science/math. We adopt this grouping for two reasons. First, aggregating majors into categories gives us sufficient statistical power for analyses that control for or examine the effects of double-major composition. Second, this aggregation scheme offers a straightforward way to compare our results with those in the prior literature.[11] The table reveals notable differences in major choice by gender. For single majors, the starkest differences can be seen in a 3:1 female preference for education majors and a 5.5:1 male preference for engineering. There is also a female preference for arts/social science and a male preference for business. The patterns for single majors generally apply to double majors, although many of the major combinations are quite rare. A large share of students who double major choose two majors in the same general field, with two arts/social science,

8. Comprehensive colleges are institutions that offer graduate education through a master's degree.

9. Table A1 in the Appendix provides a more detailed breakdown of double majoring by gender across institution types.

10. The variable "job match" is an indicator variable defined by the NSCG that takes the value of one if an individual has a "principal job closely related to highest degree," and zero otherwise.

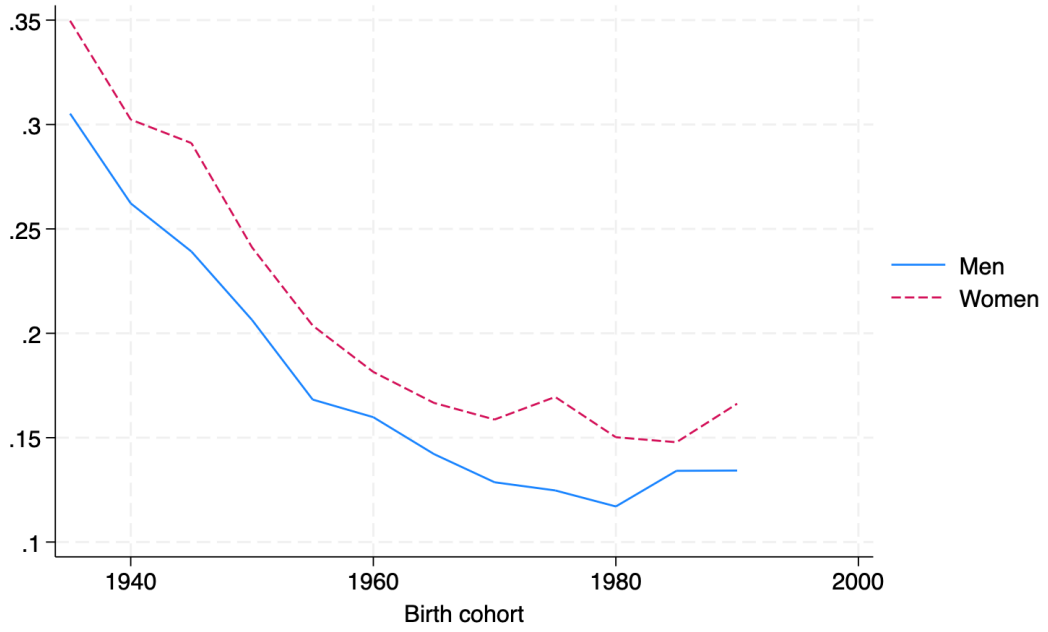
11. Del Rossi and Hersch, 2016 uses the same categories, except that it combines engineering with science/math into a single STEM category. Hemelt, 2010 does not examine specific double-major combinations, focusing instead on the effects of adding any double major to a given first major.

Table 1: Summary Statistics by Gender and Double Major Status

| | Men | | Women | |
|---|---|---|---|---|
| | Single major | Double major | Single major | Double major |
| *Control variables* | | | | |
| *Race/Ethnicity* | | | | |
| White | 0.75 | 0.76 | 0.71 | 0.75 |
| Black | 0.06 | 0.07 | 0.09 | 0.08 |
| Hispanic | 0.08 | 0.07 | 0.09 | 0.08 |
| Asian | 0.09 | 0.08 | 0.09 | 0.07 |
| American Indian | 0.00 | 0.00 | 0.00 | 0.00 |
| Pacific Islander | 0.00 | 0.00 | 0.00 | 0.00 |
| Multiple | 0.02 | 0.02 | 0.02 | 0.02 |
| Mother has BA | 0.33 | 0.32 | 0.33 | 0.32 |
| Father has BA | 0.42 | 0.40 | 0.41 | 0.40 |
| *Carnegie class.: first BA inst.* | | | | |
| Research | 0.34 | 0.31 | 0.28 | 0.27 |
| PhD-granting | 0.12 | 0.12 | 0.12 | 0.11 |
| Comprehensive | 0.27 | 0.28 | 0.32 | 0.34 |
| Liberal arts | 0.10 | 0.14 | 0.13 | 0.17 |
| Other | 0.16 | 0.14 | 0.15 | 0.12 |
| Age | 47.44 | 50.94 | 45.21 | 48.28 |
| | (14.14) | (14.89) | (13.70) | (14.55) |
| Year of first BA | 1993 | 1988 | 1995 | 1991 |
| | (14.62) | (15.38) | (13.84) | (15.00) |
| Survey year | 2017 | 2016 | 2017 | 2016 |
| | (3.62) | (3.74) | (3.59) | (3.71) |
| *Outcome variables* | | | | |
| Earnings | 85,956.80 | 83,147.41 | 46,407.32 | 46,334.43 |
| | (139,529.32) | (144,973.23) | (73,869.52) | (71,324.50) |
| Employed | 0.75 | 0.68 | 0.63 | 0.58 |
| Married | 0.74 | 0.74 | 0.68 | 0.67 |
| Number of children in household | 0.68 | 0.58 | 0.73 | 0.59 |
| | (1.07) | (1.02) | (1.07) | (0.99) |
| Good job match | 0.35 | 0.31 | 0.33 | 0.31 |
| Greater than BA | 0.33 | 0.40 | 0.33 | 0.41 |
| *N* | 240,619 | 46,212 | 191,324 | 49,444 |

"Hispanic" includes any Hispanic identification; other race/ethnicity categories are all non-Hispanic. Good job match is an indicator variable defined by the NSCG that takes a value of one if an individual has a "principal job closely related to highest degree," and zero otherwise. This variable is restricted to individuals who have no degrees higher than a single BA. Observations for this variable: 129,111, 22,800, 94,322, and 22,345.

Figure 1: Proportion of Double Majors by Cohort




two business, and two science/math double majors making up more than half of the observed combinations for both men and women.

Finally, to get a sense of the variation in our institution fixed effects, Figure A1 in the Online Appendix plots a kernel density of the distribution of individuals across institutions, with 5 percent of each tail trimmed off according to Census disclosure rules. The figure indicates that there is a wide range of institution sizes represented. Additionally, there are generally many students per institution, which facilitates making comparisons within institutions.

# 4 Results

## 4.1 Main Results

The main results using Equation 1 for all six of our outcomes are presented in Table 3.[12] In the table, each column corresponds to a different outcome. The top panel presents the estimated coefficients of interest for the "Full" sample specifications interacting the double

12. Table A2 in the Online Appendix presents results that do not control for institution fixed effects (i.e., using the public-use NSCG). Table A3 in the Online Appendix presents results using Oster (2019)'s sensitivity analysis. The broad pattern of results reported here using Diegert, Masten, and Poirier (2023)'s sensitivity analysis holds with Oster (2019)'s.

Table 2: Major Combinations by Gender

| | Men | Women |
|---|---|---|
| *Single major categories* | | |
| Arts/social science | 0.294 | 0.397 |
| Business | 0.271 | 0.188 |
| Education | 0.049 | 0.148 |
| Engineering | 0.149 | 0.027 |
| Science/math | 0.238 | 0.240 |
| *Double major categories* | | |
| Two arts/social science | 0.275 | 0.343 |
| Arts/social science and business | 0.091 | 0.079 |
| Arts/social science and education | 0.043 | 0.119 |
| Arts/social science and engineering | 0.008 | 0.002 |
| Arts/social science and science/math | 0.072 | 0.073 |
| Two business | 0.187 | 0.107 |
| Business and education | 0.008 | 0.006 |
| Business and engineering | 0.012 | 0.002 |
| Business and science/math | 0.057 | 0.027 |
| Two education | 0.033 | 0.122 |
| Education and engineering | 0.000 | 0.000 |
| Education and science/math | 0.017 | 0.026 |
| Two engineering | 0.029 | 0.004 |
| Engineering and science/math | 0.031 | 0.004 |
| Two science/math | 0.127 | 0.079 |

major indicator with gender. The subsequent panels present specifications for "Men only" and "Women only," which are our preferred specification. The bottom panel of the table provides information about the number of observations and the $R^2$ for each of the specifications.

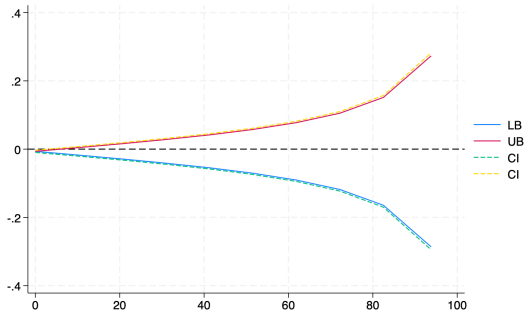
Column 1 in Table 3 indicates that, taking into account all control variables and fixed effects, there appears to be no earnings effect of double majoring for men and a statistically significant estimated differential effect of 3.4 percent between women and men. In the models

Table 3: Main Results

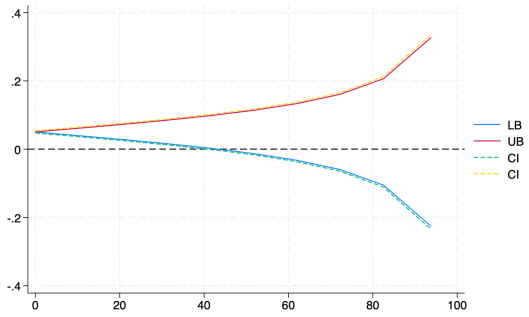
| | (1) Log Earnings | (2) Employed | (3) Married | (4) Children | (5) Job Match | (6) Grad. Ed. |
|---|---|---|---|---|---|---|
| *Full* | | | | | | |
| Double major | 0.0103 | -0.0050 | -0.0076 | 0.0079 | -0.0041 | 0.04307*** |
| | (0.01343) | (0.00509) | (0.00505) | (0.00487) | (0.00766) | (0.00528) |
| Double major × Female | 0.0337* | 0.0069 | -0.0167** | -0.0364*** | 0.0031 | 0.0155* |
| | (0.01885) | (0.00738) | (0.00711) | (0.00685) | (0.01065) | (0.00730) |
| *Men only* | | | | | | |
| Double major | -0.0059 | -0.0029 | -0.0192*** | 0.0038 | -0.0104 | 0.0369*** |
| | (0.01287) | (0.00497) | (0.00483) | (0.00476) | (0.00745) | (0.00516) |
| $\bar{r}_D^{bp}(0)$ (×100) | 6.15 | 6.79 | 39.5 | 7.91 | 19.6 | 58.7 |
| *Women only* | | | | | | |
| Double major | 0.0518*** | 0.0014 | -0.0172*** | -0.0258*** | -0.0051 | 0.0613*** |
| | (0.01345) | (0.00538) | (0.00506) | (0.00486) | (0.00750) | (0.00509) |
| $\bar{r}_D^{bp}(0)$ (×100) | 43.7 | 2.54 | 36.2 | 51.8 | 8.49 | 79.9 |
| *Full* | | | | | | |
| $N$ | 464,000 | 527,000 | 527,000 | 527,000 | 268,000 | 527,000 |
| $R^2$ | 0.2118 | 0.2088 | 0.1790 | 0.3061 | 0.1302 | 0.1266 |
| *Men only* | | | | | | |
| $N$ | 257,000 | 287,000 | 287,000 | 287,000 | 151,000 | 287,000 |
| $R^2$ | 0.2479 | 0.2707 | 0.2301 | 0.3273 | 0.1720 | 0.1696 |
| *Women only* | | | | | | |
| $N$ | 206,000 | 240,000 | 240,000 | 240,000 | 116,000 | 240,000 |
| $R^2$ | 0.1705 | 0.1893 | 0.2031 | 0.3395 | 0.1745 | 0.1566 |

FSRDC Project Number 2985; Disclosure Clearance Numbers CBDRB-FY24-P2985-R11339 and CBDRB-FY24-P2985-R11713. Results rounded according to Census disclosure rules. Each italicized row header indicates a separate regression. Job match column restricted to individuals whose highest education is a single BA given that the survey question asks about the highest degree. $\bar{r}_D^{bp}(0)$ is the selection on unobservables percentage, relative to selection on observables, required to overturn the empirical finding implied by the corresponding estimate in the table. Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

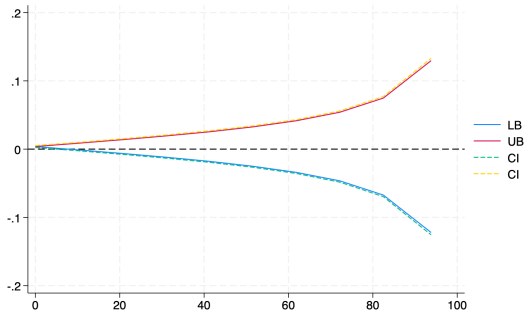
Figure 2: Identified Sets by Gender



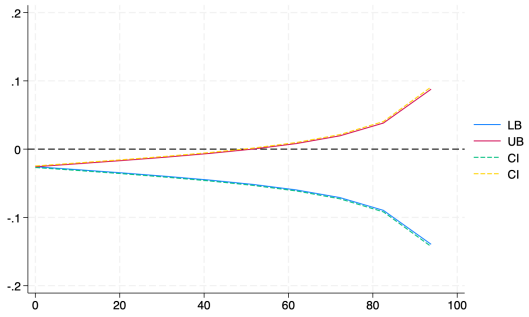

(a) Men - Log Earnings



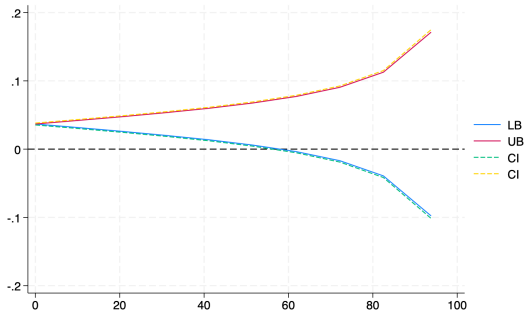

(b) Women - Log Earnings



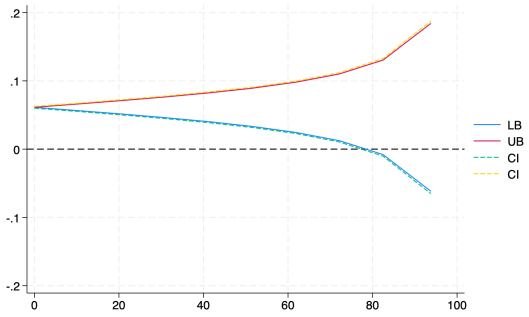

(c) Men - Children




(d) Women - Children




(e) Men - Graduate Education




(f) Women - Graduate Education

FSRDC Project Number 2985; Disclosure Clearance Number CBDRB-FY24-P2985-R11713. Results rounded according to Census disclosure rules. Figures show the identified set for the effect of a double major on each outcome and gender over the different possible percentage values of $\bar{r}_D^{bp}(0)$, the breakdown point occurs where the bounds cross the x-axis.

that allow all variables to have a differential impact by gender, women who double major have a statistically significant estimated earnings boost of 5.2 percent. When considering men only, the estimated effect of double majoring is close to zero and statistically insignificant.

The rows denoted $\bar{r}_D^{bp}(0)$ present the breakdown point based on Diegert, Masten, and Poirier (2023)'s sensitivity analysis. As explained in section 2.2, it is interpreted as the percentage of selection on unobservables relative to selection on observables that would be required to reverse our finding about the sign of the corresponding estimate. The variables included in $X_{isnat}$ that serve as benchmark for the breakdown point are parents' education and the race and ethnicity indicators. Therefore, the value $\bar{r}_D^{bp}(0) = 43.7$ in Table 3 implies that, in order for the true effect of a double major on women's earnings to be negative, the unobservables would have to be 43.7% as powerful as the impact of parents' education and the race and ethnicity indicators on the choice of double majoring. Given that these observable variables are deemed to be important in the choice to double major, and also given the other variables that have been already included in the specification (the fixed effects), it appears unlikely that there are unobservables with such explanatory power on the individual choice to double major. Therefore, we deemed this estimated effect as fairly robust to within-institution selection on unobservables into double majoring. In passing, note that, in practice, the interpretation of $\bar{r}_D^{bp}(0)$ is not the same as that of the sensitivity parameter of Oster (2019), since they represent different measures of the same sensitivity concept.

Turning to the estimated effects on employment in column 2, the results indicate null effects across samples. These findings are notable since selection into employment is a potentially significant source of gender disparity. In other words, it seems that the earnings effect of double majoring from column 1 does not operate via the extensive margin.[13] In column 3, we see that the specifications by gender point to fairly similar negative effects of double majoring on marital status. Women double majors have an effect of -1.7 and men have an effect of -1.9 percentage points. Both estimates are judged robust, with over 1/3 selection on unobservables, relative to selection on observables, required to overturn the effects.

Column 4 unveils another stark gender difference in the effect of double majoring, this time on the presence of children in the household. There appears to be no impact of double majoring on the presence of children in the household for men, but women who double major are 2.58 percentage points less likely to have them. The robustness of this estimate with respect to selection on unobservables is strong, with a $\bar{r}_D^{bp}(0)$ of 51.8.

Finally, while there is no discernible effect of double majoring on the quality of job-degree matches, as shown in column 5, column 6 indicates that there is a large effect on the propen-

13. This is reassuring, since there are approximately 63,000 instances of $0 earnings in the data, which are automatically dropped due to the log specification.

sity to attain graduate education.[14] These findings on graduate enrollment are qualitatively consistent with the results in Hemelt, 2010. The effect for men is 3.7 percentage points and strongly robust, with a $\bar{r}_D^{bp}(0)$ of 58.7. The corresponding effect for women is about 6.1 percentage points and appears extrememly robust to selection based on unobservables, with a $\bar{r}_D^{bp}(0)$ of 79.9. These findings suggest graduate education as a potential mechanism linking double majors to women's earnings, though our data preclude definitive causal identification of this channel.

The entire Diegert, Masten, and Poirier (2023) identified sets are displayed for the earnings, fertility, and graduate education outcomes for both men and women in Figure 2. These bounds tell the same story as Table 3 in visual form. $\bar{r}_D^{bp}(0)$ is determined by where the bounds first cross the x-axis. In these plots, if we were to assume that estimates are unconfounded by unobservables (i.e. $\bar{r}_D = 0$) then our point estimate would be equal to the estimated $\hat{\beta}$. Assuming unobservables are equally as powerful or more ($\bar{r}_D \geq 1$) results in a loss of identification. Any assumption in between can be accommodated by finding the size and location of the bounds at that location in the figure.

One potential concern for the interpretation of results is that gender differences in the returns to double majoring may actually reflect gender differences in the *composition* of majors. For example, perhaps women who double major tend to diversify their majors more than men who double major. In that case, differences in double major diversification could be driving gender differences in labor market outcomes, as Hanks et al., 2024 find that double majoring provides more protection against labor market shocks when the two majors are more distantly related. To assess this concern, we estimate results that disaggregate for major composition by gender. These results are shown in Tables A4 and A5 in the Appendix for men and women, respectively.[15] In the top panel of the tables, we compare double majors whose majors are in the same field ("2x –"), which we refer to as "Close," to double majors whose degrees are in different categories, which we refer to as "Far." These fields are classified using the categories shown in Table 2. Consistent with the aggregate results, we find that men who double major do not experience any earnings returns, both for those with Close and Far double majors. In contrast, women who double major experience a positive earnings return for both Close and Far double majors. These results suggest that

14. A contributing factor to this could be the smaller sample sizes that result from the way that the "job match" variable is recorded in the NSCG. Having a job-degree match is defined by the NSCG as having a "principal job closely related to highest degree." Since the question only asks about the "highest" degree, we are forced to restrict the sample for this variable to those whose highest degree is their first bachelor's degree.

15. We only obtained "sign and significance" results for these tables in order to facilitate Census disclosure review.

our findings are not driven by women diversifying their double majors, while men do not.

Even holding the closeness of the majors constant, if men and women tend to double major in different bundles of subjects, this could lead to gender differences in returns to double major to the extent that different majors and combinations of majors have different labor market returns. In the second panel of Tables A4 and A5, we disaggregate the findings to look at the returns to specific categories of double majors, using the categories defined in Table 2.[16] While we do not observe point estimates, we do see that for each combination of major categories, the earnings return for women weakly dominates the earnings return for men, signwise. These findings further support that our main findings are not driven by gender differences in major composition. While the second panel of Tables A4 and A5 looks at returns to specific combinations of double majors, it does not account for the fact that the composition of single majors, which is the omitted category, may differ between men and women. Table A6 further addresses this concern by looking at heterogeneity in results by gender when holding the comparison group of single majors constant. Each row of the table represents a separate regression. The row title reflects the base major, and coefficient estimates represent the effects of double majoring compared to single majoring in that field. For example, for the second row of estimates, Arts/Social Science, column 1 reports the earnings returns of an individual who double majors with at least one major in an Arts/Social Science field compared to Arts/Social Science single majors. Results indicate that for men, the returns to double majoring is statistically indistinguishable from zero for all fields except for a positive returns in Arts/Social Science. In contrast, for women, estimates of the returns to double majoring are positive and statistically significant in most fields, with the exception of statistically insignificant results in Engineering and Science/Math. These results provide further support that the gender differences we find in the returns to double majoring are not driven by differences in major composition.

Overall, we provide multiple supplemental analyses to assess the possibility that gender differences in the returns to double majoring are driven by differences in the composition of majors chosen by men and women. This includes assessing whether results are driven by gender differences in the propensity to diversify double majors, gender differences in the choice of majors for double majors, and gender differences in the composition of single majors. While these analyses are not definitive proof that compositional differences in majors do not influence findings, they provide considerable reassurance that this is not driving our results.

16. Unfortunately, we are underpowered to disaggregate into finer major categories.

## 4.2 Potential Mechanisms

In this section, we discuss the consistency of our findings with two potential mechanisms for explaining the estimated returns to double majoring: First, double majoring may reflect an investment in human capital (Becker, 1964; Ben-Porath, 1967; Card, 1999) or, relatedly, a method to improve a comparative advantage (Roy, 1951; Lazear, 2005; Altonji, Arcidiacono, and Maurel, 2016). Second, double majoring may function as a means of signaling ability (Spence, 1973; Weiss, 1983).[17] Human capital theory posits that double majoring leads to higher earnings through the accumulation of productive knowledge gained via education in a formal school setting. In contrast, signaling theory advances that the use of educational credentials is a way to indicate innate ability (or other unobserved traits attractive to employers). A labor market signal reveals preexisting dispositions, and to the extent that these dispositions are linked to labor market outcomes, we would expect these variables to move together (e.g., career ambition and earnings). This sets up our main contrast with the human capital and comparative advantage explanations, since in this case we would expect more uniform labor market effects that are not associated with variables outside of the labor market.[18]

What we observe in our previous results is broadly consistent with signaling theory, particularly for women. Although double majors of both genders are less likely to be married, only women are less likely to have children, conditional on age, and only women experience an earnings return to double majoring. This suggests a fertility connection with earnings for double majors. If the labor market penalizes women due to the chance of a future maternity leave, then those women who are less likely to have children or who delay childbearing due to their career focus would be incentivized to pursue costly differentiation strategies. A double major may be just such a costly and thus credible signal of intent.

The same evidence tends to cast doubt on the human capital story. Even if a double major had a unique signaling value for women, its human capital content should still be valuable for men. However, it seems that there is essentially no labor market value to a double major for men. This provides a potential answer to the debate over whether students who choose to double major have "extra" skills or lack deep knowledge. Our evidence suggests that the answer is neither. A double major bachelor's degree appears to carry, on

17. Theoretically, a double major could be viewed through additional non-mutually exclusive lenses: an affinity for an academic subject that yields non-pecuniary returns (Rosen, 1986) or a means of insuring against income risk (Hanks et al., 2024). These theories do not, in general, provide clear a priori predictions for the effect of a double major on each of our outcomes.

18. As a second order effect, an increase in the opportunity cost of time could impact these non-labor market variables, however.

average, the same human capital with potential to be rewarded in the labor market as that of a single major.

Finally, although more speculative, there is a story to be told about academic preferences *overall* as a signaling device. Double majors are significantly more likely to pursue graduate education. This makes sense since the choice to double major represents a higher level of academic achievement, as do graduate degrees. Those with stronger preferences for academics would thus naturally be more likely to have double majors and graduate degrees. At the same time, we do not see any compensating wage differentials at an aggregate level, as would be expected if preferences were for specific majors. Furthermore, the effect of graduate education for women, who have strong earnings effects, is much more stable than for men, who do not. Although we cannot rule out a similar effect for men, our evidence is at least consistent with a preference for formal education, as revealed through a double major, being an important mechanism through which women are signaling their labor market intent to employers.

# 5 Conclusion

This study provides novel evidence of the causal effects of double majoring in college on several later life outcomes. Using a combination of new data and new techniques, we show that while our top line numbers are in line with prior studies, our results differ from prior work in that we find a large gender gap in labor market returns to double majoring. Specifically, there is a notable earnings return to double majoring for women, while we find no evidence of a return for men. This effect is not driven by positive selection into the labor market (the extensive margin).

The lower propensity of women double majors to have children suggests that this effect is likely due to the signaling value that a double major provides to employers, indicating labor market ambition and thus commanding higher wages. At the same time, the gender split in returns suggests that the overall human capital associated with a double major is likely the same, on average, as that for single majors. Finally, both women and men with double majors are significantly more likely to have graduate education, presumably due to their preference for academics.

Nothing about our estimates suggests the mechanisms driving these labor market behaviors should be limited to double majoring. If women are motivated in this instance to pursue costly signaling strategies, they will likely make use of other similar devices. More research is needed to find such additional instances of gender-based signaling elsewhere in

the labor market.

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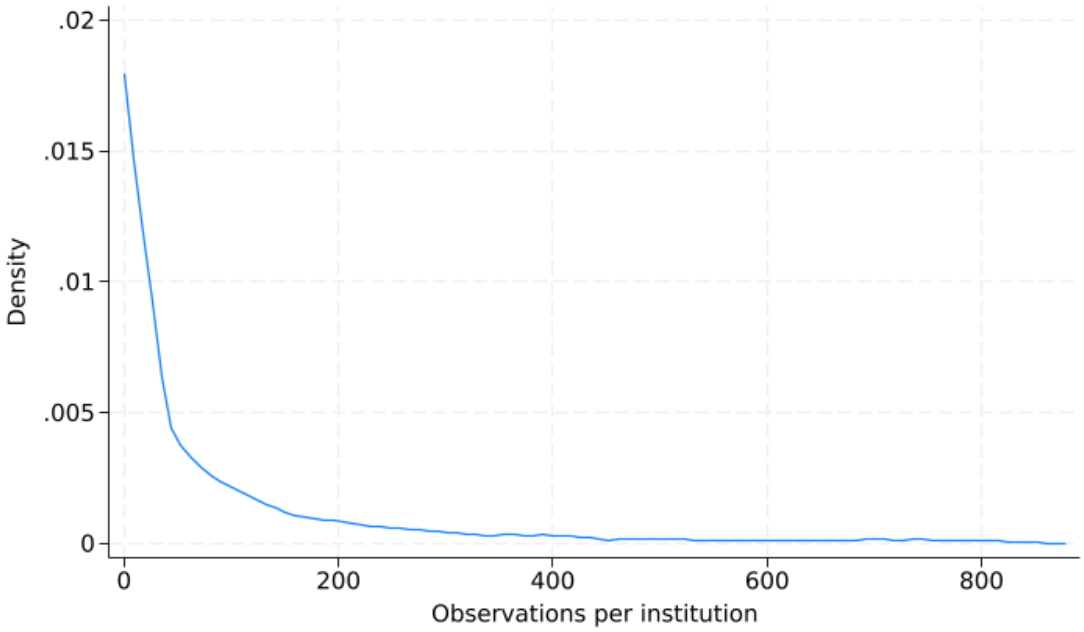
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# A Appendix

Figure A1: Distribution of Observations per Institution




FSRDC Project Number 2985; Disclosure Clearance Number CBDRB-FY24-P2985-R11339. Data trimmed according to Census disclosure rules. Default bandwidth used.

Table A1: Double Majors by Carnegie Category

| | Men | Women |
|---|---|---|
| Research University I | 0.150 | 0.174 |
| Research University II | 0.151 | 0.182 |
| Doctorate Granting I | 0.165 | 0.182 |
| Doctorate Granting II | 0.150 | 0.151 |
| Comprehensive I | 0.165 | 0.194 |
| Comprehensive II | 0.148 | 0.179 |
| Liberal Arts I | 0.234 | 0.275 |
| Liberal Arts II | 0.196 | 0.193 |
| Other | 0.138 | 0.156 |

The table displays the propensity for students to double major separately by institution type and gender. Institution types are defined based on 1994 Carnegie classifications. Research Universities award at least 50 doctoral degrees, and Research Universities I receive at least $40 million dollars in federal funding while Research Universities II receive between $15-$40 million. Doctoral Universities I award at least 40 doctoral degrees in at least 5 fields, while Doctoral II Universities award at least 10 doctoral degrees in at least 3 fields. Comprehensive I and II institutions are both Master's degree-granting institutions, with Comprehensive I schools granting a higher volume of degrees than Comprehensive II schools. Liberal Arts colleges are primarily undergraduate institutions, with Liberal Arts I schools awarding at least 40 percent of degrees in liberal arts fields and having more selective admissions, and Liberal Arts II schools awarding less than 40 percent of degrees in liberal arts fields and less selective admissions.

Table A2: Results not Controlling for Institution Fixed Effects

| | (1) Log Earnings | (2) Employed | (3) Married | (4) Children | (5) Job Match | (6) Grad. Ed. |
|---|---|---|---|---|---|---|
| *Full* | | | | | | |
| Double major | 0.00328 | -0.00802 | -0.0052 | 0.00927 | -0.00995 | 0.0537*** |
| | (0.817) | (0.103) | (0.005) | (0.005) | (0.206) | (0.000) |
| Double major × female | 0.0423* | 0.00731 | -0.0143 | -0.0393*** | 0.00927 | 0.0106 |
| | (0.032) | (0.319) | (0.007) | (0.007) | (0.405) | (0.174) |
| *Men only* | | | | | | |
| Double major | -0.00536 | -0.00553 | -.0171 | 0.0035 | -0.00661 | 0.0504*** |
| | (0.702) | (0.261) | (0.005) | (0.005) | (0.402) | (0.000) |
| $\bar{r}_D^{bp}(0)$ (×100) | 4.81 | 12.3 | 33.9 | 6.82 | 12.7 | 67.9 |
| *Women only* | | | | | | |
| Double major | 0.0601*** | -0.000719 | -0.0101 | -0.0253*** | -0.00287 | 0.0668*** |
| | (0.000) | (0.896) | (0.005) | (0.005) | (0.719) | (0.000) |
| $\bar{r}_D^{bp}(0)$ (×100) | 46.2 | 1.88 | 22.5 | 51.6 | 6.13 | 82.4 |
| *Full* | | | | | | |
| $N$ | 464145 | 527596 | 527596 | 527596 | 268577 | 527596 |
| $R^2$ | 0.1528 | 0.1850 | 0.1188 | 0.2605 | 0.0471 | 0.0489 |
| *Men only* | | | | | | |
| $N$ | 257446 | 286829 | 286829 | 286829 | 151910 | 286829 |
| $R^2$ | 0.1522 | 0.2340 | 0.1453 | 0.2571 | 0.0569 | 0.0577 |
| *Women only* | | | | | | |
| $N$ | 206699 | 240766 | 240766 | 240766 | 116667 | 240766 |
| $R^2$ | 0.0809 | 0.1331 | 0.1109 | 0.2703 | 0.0450 | 0.0497 |

Each italicized row header indicates a separate regression. Job match column restricted to individuals whose highest education is a single BA. $\bar{r}^{bp}(0)$ is the (absolute value) selection percentage required for the true effect to have a different sign from what we estimate. Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A3: Main Results with Oster Sensitivity Parameters

| | (1) Log Earnings | (2) Employed | (3) Married | (4) Children | (5) Job Match | (6) Grad. Ed. |
|---|---|---|---|---|---|---|
| *Full* | | | | | | |
| Double major | 0.0103 | -0.0050 | -0.0076 | 0.0079 | -0.0041 | 0.04307*** |
| | (0.01343) | (0.00509) | (0.00505) | (0.00487) | (0.00766) | (0.00528) |
| Double major × Female | 0.0337* | 0.0069 | -0.0167** | -0.0364*** | 0.0031 | 0.0155* |
| | (0.01885) | (0.00738) | (0.00711) | (0.00685) | (0.01065) | (0.00730) |
| *Men only* | | | | | | |
| Double major | -0.0059 | -0.0029 | -0.0192*** | 0.0038 | -0.0104 | 0.0369*** |
| | (0.01287) | (0.00497) | (0.00483) | (0.00476) | (0.00745) | (0.00516) |
| $\beta_{Oster}$ | -0.0478 | 0.0129 | -0.0206 | 0.0180 | 0.0178 | -0.0082 |
| $\delta_{Oster}$ | 5.980 | 0.487 | 30.810 | 0.038 | 0.156 | 0.788 |
| *Women only* | | | | | | |
| Double major | 0.0518*** | 0.0014 | -0.0172*** | -0.0258*** | -0.0051 | 0.0613*** |
| | (0.01345) | (0.00538) | (0.00506) | (0.00486) | (0.00750) | (0.00509) |
| $\beta_{Oster}$ | 0.0204 | 0.0541 | -0.0270 | -0.0193 | 0.0087 | 0.0608 |
| $\delta_{Oster}$ | 3.000 | 1.221 | 29.740 | 1.899 | 0.279 | 2.682 |
| *Full* | | | | | | |
| $N$ | 464,000 | 527,000 | 527,000 | 527,000 | 268,000 | 527,000 |
| $R^2$ | 0.2118 | 0.2088 | 0.1790 | 0.3061 | 0.1302 | 0.1266 |
| *Men only* | | | | | | |
| $N$ | 257,000 | 287,000 | 287,000 | 287,000 | 151,000 | 287,000 |
| $R^2$ | 0.2479 | 0.2707 | 0.2301 | 0.3273 | 0.1720 | 0.1696 |
| *Women only* | | | | | | |
| $N$ | 206,000 | 240,000 | 240,000 | 240,000 | 116,000 | 240,000 |
| $R^2$ | 0.1705 | 0.1893 | 0.2031 | 0.3395 | 0.1745 | 0.1566 |

FSRDC Project Number 2985; Disclosure Clearance Number CBDRB-FY24-P2985-R11339. Results rounded according to Census disclosure rules. Job match column restricted to individuals whose highest education is a single BA. $\beta_{Oster}$ is the bound on the double major coefficient assuming "equal selection," i.e. a selection ratio of 1 between unobservables and observables. $\delta_{Oster}$ is the selection ratio required for the estimated effect to become zero. As a rule-of-thumb, a selection ratio greater than 1 indicates that the estimate is robust to selection based on unobservables. See Oster (2019) for details. Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

In Tables A4 and A5, corresponding to men and women, we explore additional heterogeneity in double major effects for each outcome. We only obtained "sign and significance" results for these tables to facilitate Census disclosure review. Table A4 displays heterogeneity results for men and Table A5 displays results for women. First, we compare double majors whose majors are in the same field ("2x –"), which we refer to as "Close," to double majors whose degrees are in different categories, which we refer to as "Far." These fields are classified using the categories shown in Table 2. The only apparent difference between genders is that men are less likely to be employed when they have a dissimilar major combination and women are less likely to be married when they have a similar combination.[19]

Next, we look at the major categories themselves. The most striking feature here is the similarity (in sign and significance) between men and women who double major with a double business degree. This is one of the few instances where employment, children, and job match indicators are all positive for men who double major, and it is the only instance where earnings, employment, job match, and graduate education effects are the same across genders. The effect on having children is positive for men and negative for women, but it is similarly negative for both genders on graduate education. This suggests that the proposed signaling effect of a double major for women need not operate exclusively via an academic channel. Women with a double business degree seem to be less academically inclined than other women with double majors, but are still receiving an earnings boost.

Other cases where genders show some similarity by categories are positive earnings effects for engineering and science/math combinations, negative employment effects for double education and education/arts and science combinations, and negative effects on having children and on job match for combinations of double major involving arts and science degrees. Business and science/math combinations, like for double business majors, have opposite effects on having children, with men more likely to have children and women less likely to have them. Overall, there are no clear patterns here, suggesting that aggregate effects are unlikely to be driven by a student's choice of majors.

Finally, we look at several categories of institutional selectivity. Earnings returns to double majoring for men are only positive for Ivy Plus graduates, while earnings effects for women are only positive for non-Ivy Plus. This pattern is similar across SAT and admissions rate quintiles, with effects for men positive at the most selective schools and effects for women positive at either end of the selectivity ranking.[20] For women, there is also a negative effect

19. Thus, we do not have enough evidence to assess the argument that double majors have a labor market insurance value.

20. The most selective schools are in the 1$^{st}$ admission rate quintile.

of double majoring on the choice to have children at either end of the admission rate ranking, but not for average SAT scores. There is no particular pattern by institution selectivity for the other outcomes.

We also explored heterogeneity by race and ethnicity, shown in Table A7, and by major composition, shown in Table A6. A priori, signaling might be seen as valuable for racial and ethnic minorities in a manner similar to women, but this analysis suggests that if this is occurring, it does not rise to a level that we can detect in our data.[21] Similarly, while it is plausible that the composition of majors—rather than the fact of double majoring—might be driving our results, we find little evidence supporting this hypothesis.

21. Since non-White groups represent a minority of the overall sample, we cannot rule out the possibility that a lack of significant results is driven by a lack of statistical power.

Table A4: Double Major Heterogeneity (Men) - Sign and Significance

| | (1) Log Earnings | (2) Employed | (3) Married | (4) Children | (5) Job Match | (6) Grad. Ed. |
|---|---|---|---|---|---|---|
| *Major similarity* | | | | | | |
| Close | ns | ns | -** | ns | ns | +*** |
| Far | ns | -** | -** | ns | ns | +* |
| *Major category combination* | | | | | | |
| 2x Art/Sci | -*** | -* | -*** | -*** | -*** | +*** |
| Art/Sci + Bus | +* | ns | -** | ns | -*** | -* |
| Art/Sci + Edu | -*** | -*** | ns | ns | ns | +*** |
| Art/Sci + Engr | ns | ns | ns | ns | ns | -** |
| Art/Sci + Sci/Math | ns | ns | -*** | ns | ns | +*** |
| 2x Bus | +*** | +*** | ns | +** | +** | -*** |
| Bus + Edu | -* | ns | ns | ns | ns | ns |
| Bus + Engr | +*** | ns | ns | ns | ns | ns |
| Bus + Sci/Math | +*** | ns | +* | +*** | +** | -** |
| 2x Edu | -*** | -*** | ns | ns | +*** | +*** |
| Edu + Engr | ns | ns | ns | ns | ns | ns |
| Edu + Sci/Math | -** | -** | ns | ns | ns | +*** |
| 2x Engr | +*** | ns | ns | ns | +*** | ns |
| Engr + Sci/Math | +*** | ns | ns | ns | ns | -* |
| 2x Sci/Math | +*** | +* | ns | +*** | +*** | +*** |
| *Ivy league* | | | | | | |
| Non-Ivy Plus | ns | ns | -*** | ns | ns | +*** |
| Ivy Plus | +** | ns | -** | -* | ns | +* |
| *Average SAT quintile* | | | | | | |
| 1st | ns | -* | -*** | -** | ns | +*** |
| 2nd | ns | ns | ns | ns | ns | ns |
| 3rd | -* | ns | -* | ns | ns | +*** |
| 4th | ns | ns | -* | ns | -** | +*** |
| 5th | +** | ns | ns | ns | ns | ns |
| *Admit rate quintile* | | | | | | |
| 1st | +*** | +* | -* | ns | ns | +*** |
| 2nd | ns | ns | ns | ns | ns | +** |
| 3rd | ns | ns | -** | ns | ns | +*** |
| 4th | ns | ns | ns | ns | ns | +* |
| 5th | ns | -* | -*** | ns | ns | +*** |

FSRDC Project Number 2985; Disclosure Clearance Number CBDRB-FY24-P2985-R11338. Each italicized row header indicates a separate regression. Quintiles are ordered numerically, from smallest to greatest (the greatest average SAT score is in the 5th quintile and the lowest average admissions rate is in the 1st quintile). Double majors in which both degrees are in the same subject category (see Table 2) are classified as "Close," while double majors in which the two degrees are in different subject categories are classified as "Far." * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, *ns* $p \geq 0.05$

Table A5: Double Major Heterogeneity (Women) - Sign and Significance

| | (1) Log Earnings | (2) Employed | (3) Married | (4) Children | (5) Job Match | (6) Grad. Ed. |
|---|---|---|---|---|---|---|
| *Major similarity* | | | | | | |
| Close | +** | ns | -*** | -*** | ns | +*** |
| Far | +* | ns | ns | -* | ns | +*** |
| *Major category combination* | | | | | | |
| 2x Art/Sci | ns | ns | -*** | -*** | -*** | +*** |
| Art/Sci + Bus | +* | ns | -* | -*** | -* | -** |
| Art/Sci + Edu | ns | -*** | +* | ns | ns | +*** |
| Art/Sci + Engr | ns | +** | ns | -** | ns | ns |
| Art/Sci + Sci/Math | +* | ns | -* | ns | ns | +*** |
| 2x Bus | +*** | +*** | ns | -** | +* | -*** |
| Bus + Edu | ns | -** | ns | ns | -*** | ns |
| Bus + Engr | +*** | ns | +* | ns | ns | ns |
| Bus + Sci/Math | +*** | +*** | ns | +** | ns | ns |
| 2x Edu | ns | -*** | ns | ns | +*** | +*** |
| Edu + Engr | +*** | ns | -*** | -* | -** | ns |
| Edu + Sci/Math | ns | ns | ns | ns | +*** | +*** |
| 2x Engr | +*** | ns | ns | ns | ns | +* |
| Engr + Sci/Math | +*** | +** | +* | -* | ns | +* |
| 2x Sci/Math | +*** | +*** | -* | -** | +* | +*** |
| *Ivy league* | | | | | | |
| Non-Ivy Plus | +*** | ns | -** | -*** | ns | +*** |
| Ivy Plus | ns | ns | ns | -* | ns | ns |
| *Average SAT quintile* | | | | | | |
| 1st | +* | ns | ns | ns | +* | +*** |
| 2nd | ns | ns | ns | -*** | ns | +*** |
| 3rd | ns | ns | -* | ns | ns | +*** |
| 4th | +** | ns | ns | -* | ns | +*** |
| 5th | +*** | ns | ns | ns | ns | +*** |
| *Admit rate quintile* | | | | | | |
| 1st | +*** | +* | ns | -** | ns | +*** |
| 2nd | ns | ns | -*** | ns | ns | +*** |
| 3rd | ns | ns | ns | ns | ns | +*** |
| 4th | ns | -** | ns | -** | ns | +*** |
| 5th | +* | ns | ns | -*** | ns | +*** |

FSRDC Project Number 2985; Disclosure Clearance Number CBDRB-FY24-P2985-R11338. Each italicized row header indicates a separate regression. Quintiles are ordered numerically, from smallest to greatest (the greatest average SAT score is in the 5th quintile and the lowest average admissions rate is in the 1st quintile). Double majors in which both degrees are in the same subject category (see Table 2) are classified as "Close," while double majors in which the two degrees are in different subject categories are classified as "Far." * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, *ns* $p \geq 0.05$

Table A6: Double Major Results by Major Composition - Sign and Significance

| | (1) Log Earnings | (2) Employed | (3) Married | (4) Children | (5) Job Match | (6) Grad. Ed. |
|---|---|---|---|---|---|---|
| *Men* | | | | | | |
| Pooled Categories | ns | ns | -*** | ns | ns | +*** |
| Arts/Social Science | +*** | ns | ns | ns | +** | +*** |
| Business | ns | ns | -** | ns | +** | +** |
| Education | ns | ns | ns | ns | ns | ns |
| Engineering | ns | -** | -*** | ns | ns | -* |
| Science/Math | ns | ns | -* | +** | -*** | +** |
| *Women* | | | | | | |
| Pooled Categories | +*** | +* | -*** | -*** | -* | +*** |
| Arts/Social Science | +*** | ns | ns | -* | +*** | +*** |
| Business | +*** | +** | ns | -* | +* | +** |
| Education | +** | ns | ns | ns | ns | +** |
| Engineering | ns | ns | ns | -** | ns | ns |
| Science/Math | ns | ns | -*** | -* | -*** | +*** |

FSRDC Project Number 2985; Disclosure Clearance Numbers CBDRB-FY24-P2985-R11554 and CBDRB-FY24-P2985-R11595. Unique to this table: separate regressions, with separate sample definitions, are shown in *each* line. Each major category is defined by an individual having *at least one* major of the relevant type. "Pooled category" regressions control for each category dummy (note that these are not mutually exclusive for double majors). Category specific regressions are restricted to that category (note that each specific regression's sample significantly overlaps with the others' due to the presence of double majors). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, *ns* $p \geq 0.05$

Table A7: Double Major Results by Race and Ethnicity - Sign and Significance

| | (1) Log Earnings | (2) Employed | (3) Married | (4) Children | (5) Job Match | (6) Grad. Ed. |
|---|---|---|---|---|---|---|
| *Men* | | | | | | |
| Asian | ns | ns | ns | ns | ns | -*** |
| Native American | +* | ns | ns | ns | ns | ns |
| Black | ns | ns | ns | ns | ns | -*** |
| Hispanic | ns | ns | ns | +* | ns | ns |
| Pacific Islander | ns | ns | ns | ns | +* | ns |
| Multiple | ns | ns | ns | ns | ns | ns |
| *Women* | | | | | | |
| Asian | ns | ns | ns | ns | ns | -*** |
| Native American | ns | ns | ns | ns | ns | ns |
| Black | ns | ns | ns | ns | ns | -*** |
| Hispanic | +** | ns | ns | ns | ns | ns |
| Pacific Islander | ns | ns | ns | ns | ns | ns |
| Multiple | ns | ns | ns | ns | ns | ns |

FSRDC Project Number 2985; Disclosure Clearance Number CBDRB-FY24-P2985-R11536. Each italicized row header indicates a separate regression. "Hispanic" is defined as any race with Hispanic ethnicity; other racial categories are non-Hispanic only. Omitted category is "White." * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, *ns* $p \geq 0.05$